

Week 1 Assignment: Welcome To The World Of Conceptual Thinking

Assignment Description

Part 1: Conceptual Thinking in Practice (Group Assignment)

Task:

In groups of 4, begin the **Planning** and **High-Level Analysis** phases of a software solution for a real-world problem on campus or in everyday life. Use **Conceptual Thinking principles** and follow the **Planning** and **Analysis** phases of the **Software Development Life Cycle (SDLC)** to define your solution. Use: PowerPoints + internet

Steps:

1. Choose a Problem:

- Identify a common problem students face on campus or a real-world problem that could be solved using a software solution.
- **Examples:**
 - Finding available study rooms.
 - Tracking assignment deadlines.
 - Managing meal plans.
 - Scheduling group study sessions.

2. Planning Phase:

- **Problem Statement:** Clearly describe the problem you are solving. What is the issue, and why does it need a solution?
- **Goal:** Define the goal of your software. What is the main purpose of the software? What should it help users achieve?
- **Stakeholders:** Identify the primary users of the software (students, professors, administration, etc.).
- **Project Scope:** Define the scope of the project—what features will be included, and what will be left out at this stage?

3. High-Level Analysis:

- **Functional Requirements:** List the key features your software should have. What must the software be able to do?
 - **Example Features:**
 - Search for available rooms.
 - Book rooms for a specific time.
 - Notify students of upcoming reservations.
- **Non-Functional Requirements:** Consider aspects like performance, security, and scalability.
 - **Example:** The system should handle up to 100 students searching for rooms at the same time.

Part 2: Theory and Research Questions (Individual Research)

Task:

Individually, research the following Conceptual Thinking and SDLC-related questions. Write a **short summary (200-300 words) for each question**, providing clear explanations and examples where appropriate.

Questions:

1. **What is Conceptual Thinking?**
 - Provide a definition of Conceptual Thinking and explain why it is important in today's world. Include real-world examples of software systems that rely on good Conceptual Thinking practices.
2. **What are the phases of the Software Development Life Cycle (SDLC)?**
 - Describe the phases of the SDLC and explain the role of the **Planning** and **Analysis** phases in software development. Why are these phases critical to the success of a software project?
3. **Waterfall vs. Iterative Development:**
 - Compare the **Waterfall** and **Iterative Development** models of the SDLC. Explain the strengths and weaknesses of each model, and give an example of a project that would benefit from each approach.
4. **Research Company Methodologies:**
 - Research at least two companies to investigate which **Software Development methodologies** they use (e.g., Agile, Waterfall, Scrum, Iterative Development). If possible, reach out to the companies or explore their websites, blogs, or interviews to find more information.
 - **Question:** How does the methodology they use impact their software development process, and why do you think they have chosen this particular method?

Grading Criteria:

1. **Group Deliverable:**
 - Clear problem statement and goal.
 - Well-defined functional and non-functional requirements.
 - Evidence of planning and understanding of the Analysis phase.
2. **Individual Research Deliverable:**
 - Quality of research.
 - Clear and concise explanations.
 - Understanding of key Conceptual Thinking concepts, SDLC models, and methodologies used by companies.